

Introduction to Safety

Concept of Safety, Importance of industrial safety, Safety scenario in Indian mines

Accident Analysis and Control

Concept of an accident, accident classification, Causes of accidents; Domino sequence, Unsafe acts and conditions; Safety performance measures

Documentation of accidents, Accident investigation and analysis; Role of supervisory staff and safety committee for accident prevention.

Hazard, Risk Issues and Hazard Assessment

Concept of hazard, hazard monitoring, assessment; Risk management, Stochastic models for accident prediction, Consequence modeling

Preliminary hazard analysis (PHA), Hazard operability studies (HAZOP), Safety warning systems.

Safety Management Plan

Behavioural Safety

Human behavioural study; Human error analysis

Safety Audit

Components of safety audit, Audit methodology, Non conformity reporting (NCR), Audit Checklist and report.

Safety Management

Principles of safety management – Workplace design, Development of safety culture, Safety Barrier; Promoting Safety at workplace

Systems Safety

Systems engineering approach to safety management.